Alec Kriebel
Independent Researcher
me@aleckriebel.com

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Editors
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Dear Editors,

Please consider the manuscript "Strong Tree-Childness Is a Sharp Generic-Identifiability Boundary for Level-2 Jukes–Cantor Networks". It establishes a complete generic observational-equivalence theorem for a broad class of phylogenetic network models and a sharp counterboundary immediately outside that class.

For binary, already-simple, strongly tree-child level-2 semi-directed networks, a source-relative full-dimensional regular containment under the open four-state Jukes–Cantor model occurs exactly when the labelled bridge trees agree and corresponding blobs are isomorphic or differ by ordinary triangle redirection. Thus there are no proper one-sided generic containments. The proof combines Fourier flattening ranks, an exact incidence-scaling bridge quotient, semialgebraic localization, and a finite graph-to-polynomial classification promoted to arbitrary subdivisions. Explicit weakly-but-not-strongly tree-child families show the hypothesis is sharp for every number of taxa at least four.

The work lies at the intersection of mathematical biology, algebraic statistics, graph theory, and real algebraic geometry. Exact primary and separately implemented replay certificates, mutation tests, and a deterministic archive support the load-bearing finite computations.

Generative AI systems assisted with exploratory algebra, implementation, drafting, and adversarial review. Earlier automated claims that failed audit were rejected and quarantined. The manuscript contains a full disclosure; I accept responsibility for all claims, code, and submission. No human specialist review is claimed.

I confirm, subject to the final checks recorded in the accompanying human checklist, that this manuscript is original, is not under consideration elsewhere, and has one author. No specific funding supported the work, and I have no competing interests relevant to it.

Thank you for your consideration.

Sincerely,

Alec Kriebel
Independent Researcher
me@aleckriebel.com